$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

$\begin{pmatrix}1\\1\end{pmatrix} \begin{pmatrix}2\\1\end{pmatrix} \begin{pmatrix}1\\2\end{pmatrix} \begin{pmatrix}5\\4\end{pmatrix} \begin{pmatrix}6\\5\end{pmatrix} \begin{pmatrix}6\\6\end{pmatrix}$

$C_1 \; C_1 \; C_1 \; C_2 \; C_2 \; C_2$

Initial $\underline{W} = \begin{pmatrix}4\\-2\\-2\end{pmatrix}$ *Step η = 2*

C_1 positive C_2 negative Zero is unaccepted

Check classes of $\begin{pmatrix}0\\0\end{pmatrix}_{Z_1} \begin{pmatrix}7\\6\end{pmatrix}_{Z_1} \begin{pmatrix}3\\2\end{pmatrix}_{Z_1}$

Ugmented Vectors (from Top)

$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

$\begin{pmatrix}1\\1\\1\end{pmatrix} \begin{pmatrix}1\\2\\1\end{pmatrix} \begin{pmatrix}1\\1\\2\end{pmatrix} \begin{pmatrix}1\\5\\4\end{pmatrix} \begin{pmatrix}1\\6\\5\end{pmatrix} \begin{pmatrix}1\\6\\6\end{pmatrix}$

$C_1 \; C_1 \; C_1 \; C_2 \; C_2 \; C_2$

Iteration 1

$d(x_1) = \begin{pmatrix}1&1&1\end{pmatrix}\begin{pmatrix}4\\-2\\-2\end{pmatrix} = 0$ ⇒ Update Weight $\underline{W} = \begin{pmatrix}4\\-2\\-2\end{pmatrix} + 2 * \begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}6\\0\\0\end{pmatrix}$ ⇐

$d(x_2) = \begin{pmatrix}1&2&1\end{pmatrix}\begin{pmatrix}6\\0\\0\end{pmatrix} = 6$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}6\\0\\0\end{pmatrix}$

$d(x_3) = \begin{pmatrix}1&1&2\end{pmatrix}\begin{pmatrix}6\\0\\0\end{pmatrix} = 6$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}6\\0\\0\end{pmatrix}$

$d(x_4) = \begin{pmatrix}1&5&4\end{pmatrix}\begin{pmatrix}6\\0\\0\end{pmatrix} = 6$ ⇒ Update Weight $\underline{W} = \begin{pmatrix}6\\0\\0\end{pmatrix} - 2 * \begin{pmatrix}1\\5\\4\end{pmatrix} = \begin{pmatrix}4\\-10\\-8\end{pmatrix}$ ⇐

$d(x_5) = \begin{pmatrix}1&6&5\end{pmatrix}\begin{pmatrix}4\\-10\\-8\end{pmatrix} = -96$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}4\\-10\\-8\end{pmatrix}$

$d(x_6) = \begin{pmatrix}1&6&6\end{pmatrix}\begin{pmatrix}4\\-10\\-8\end{pmatrix} = -104$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}4\\-10\\-8\end{pmatrix}$